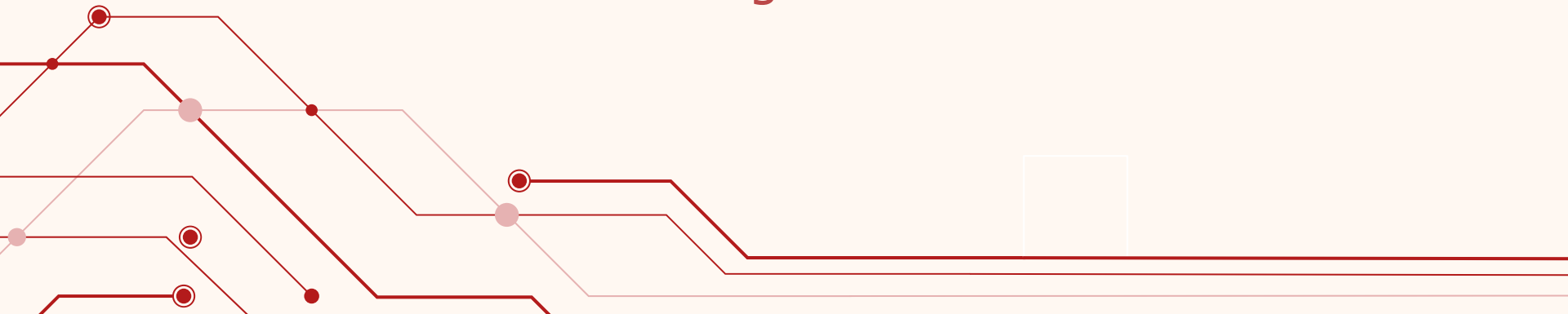


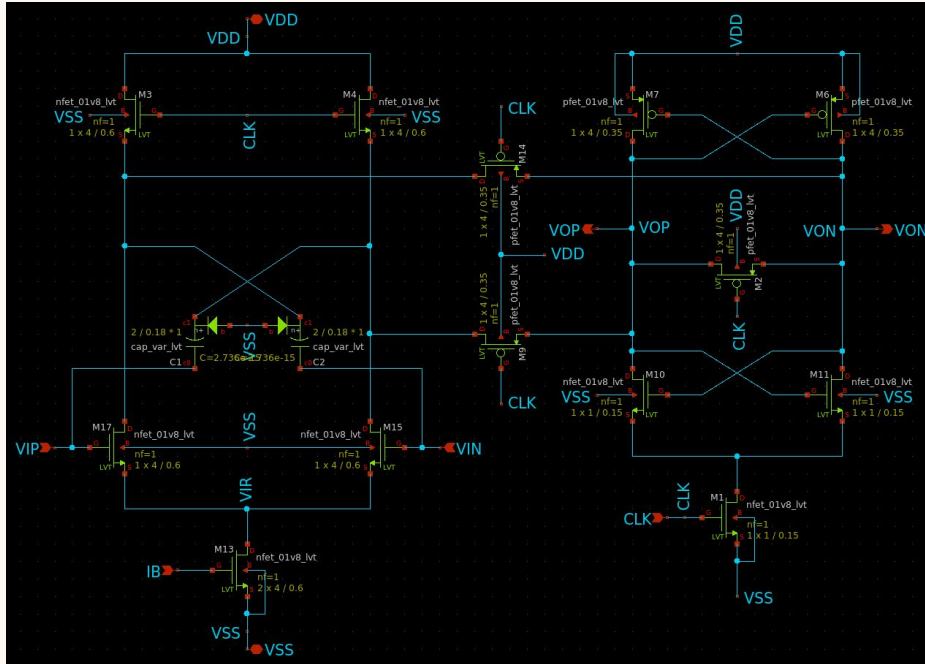
Analog Design

Caden Xu • Daniel Kaminski
David Bertuch • Jon Ho • Taylor Do • Thomas
Figura

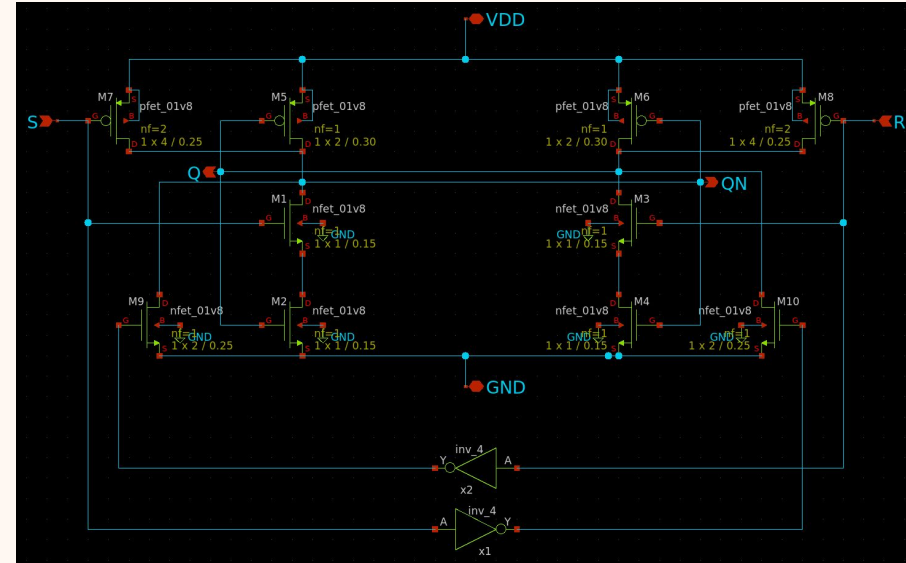




Flash ADC - Top Level Design

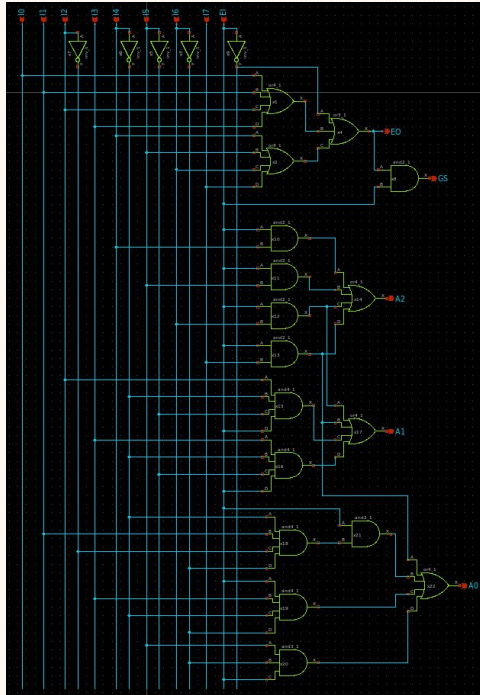


Class AB Comparator

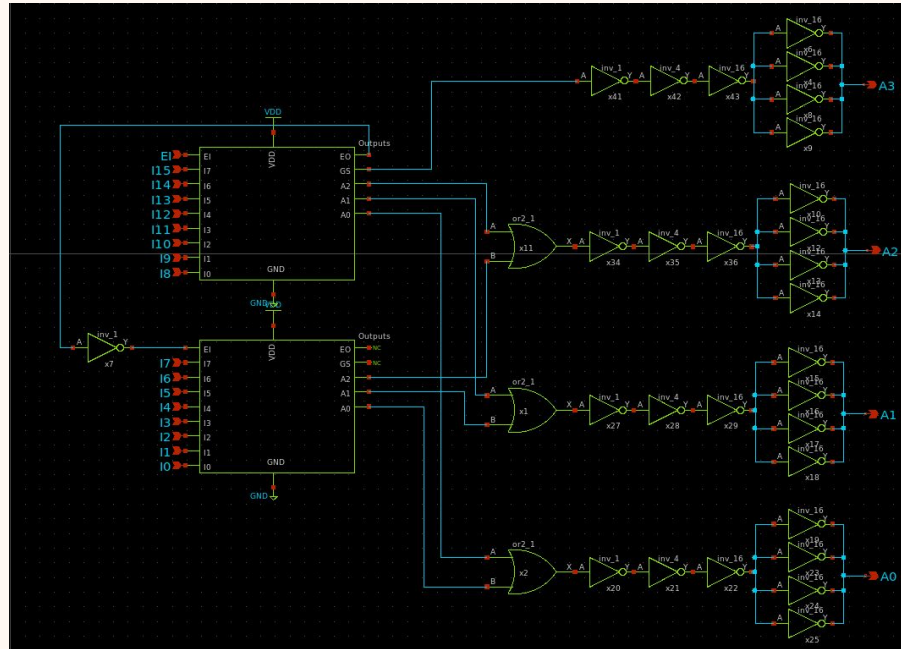


RS Latch

Flash ADC - Digital Circuitry

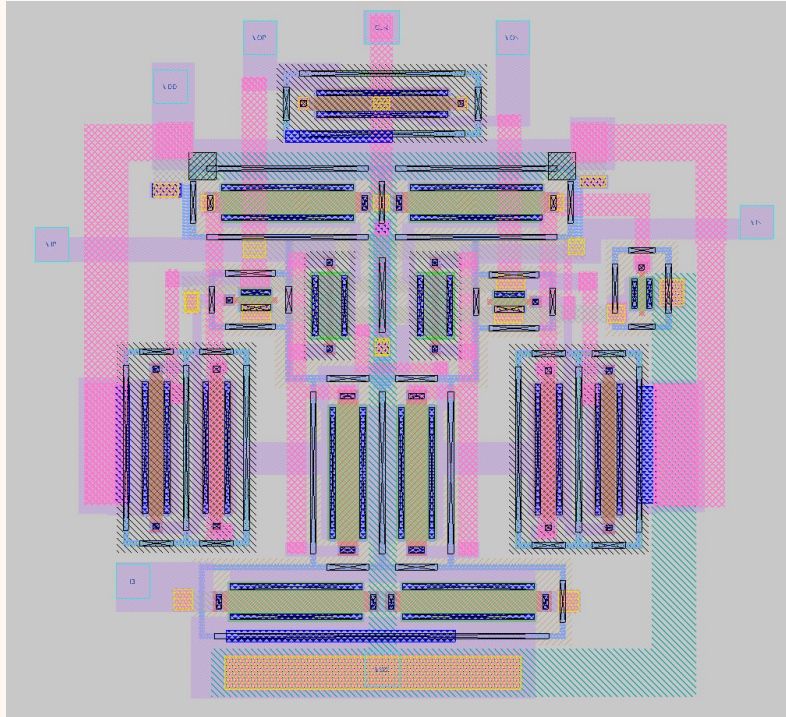


8 to 3 Priority Encoder

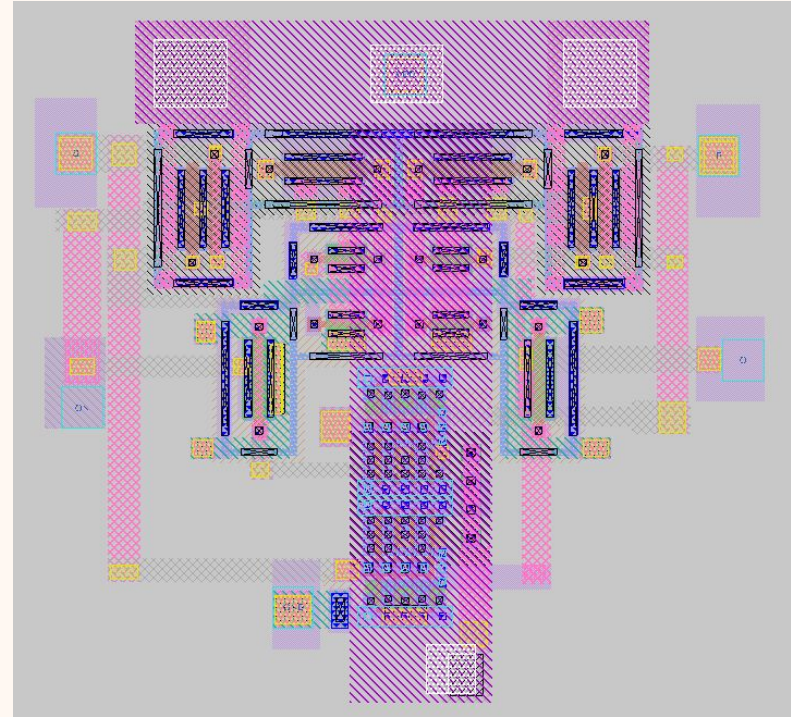


16 to 4 Priority Encoder

Flash ADC – Layout

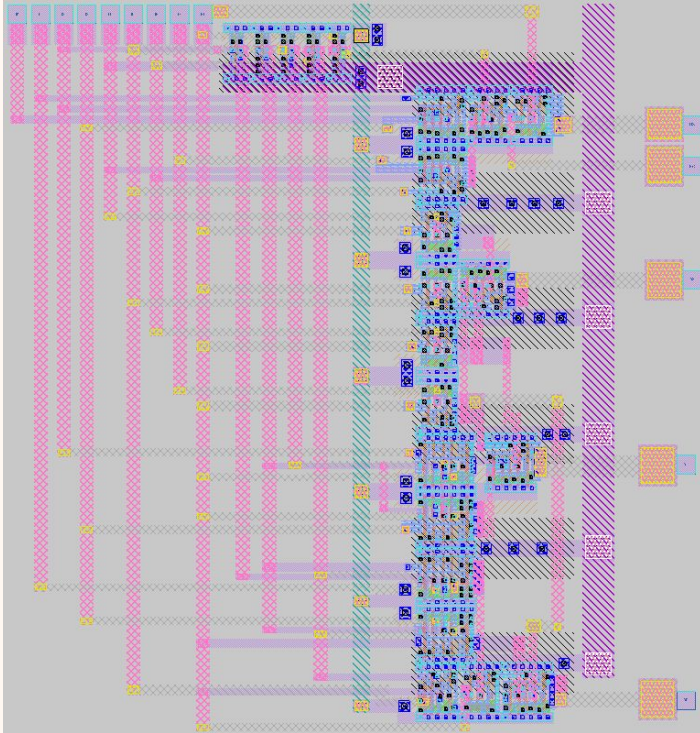


Comparator Layout

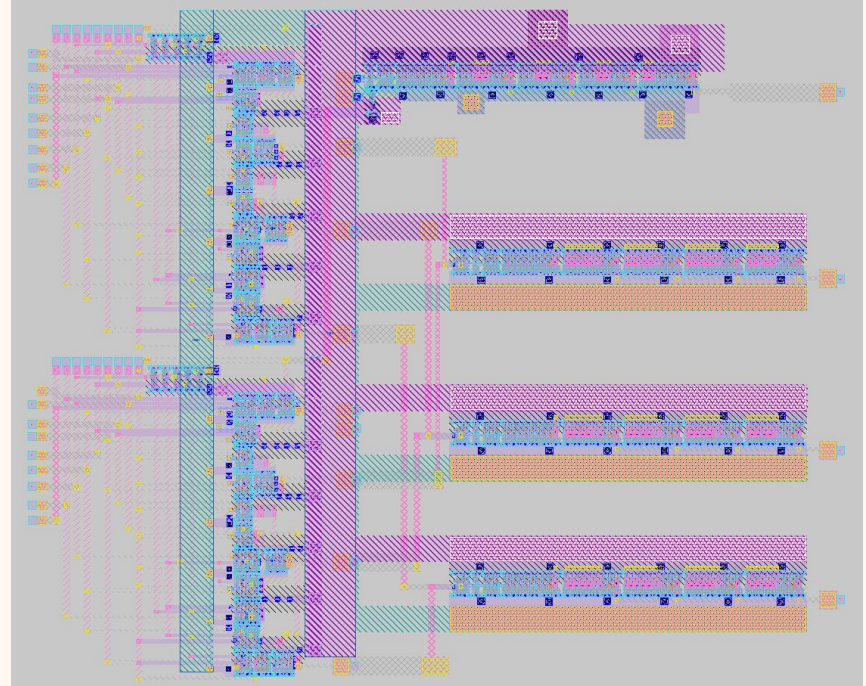


RS Latch

Flash ADC - Layout



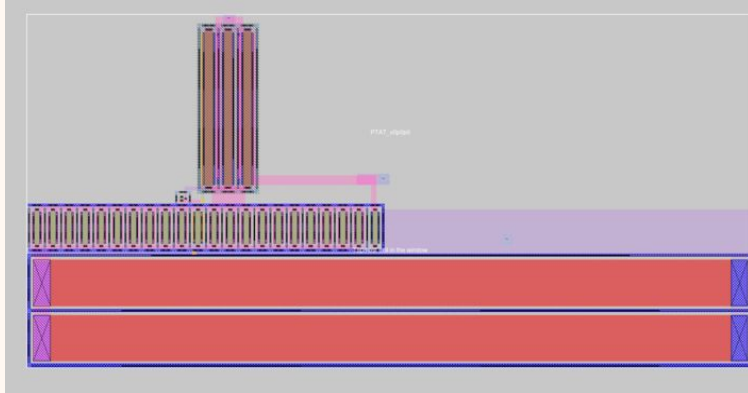
8 to 3 Priority Encoder



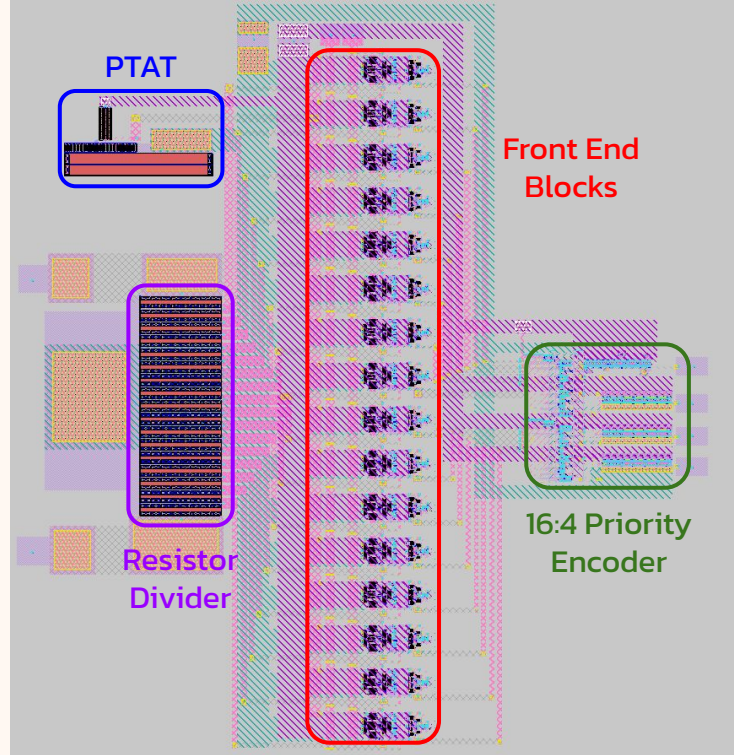
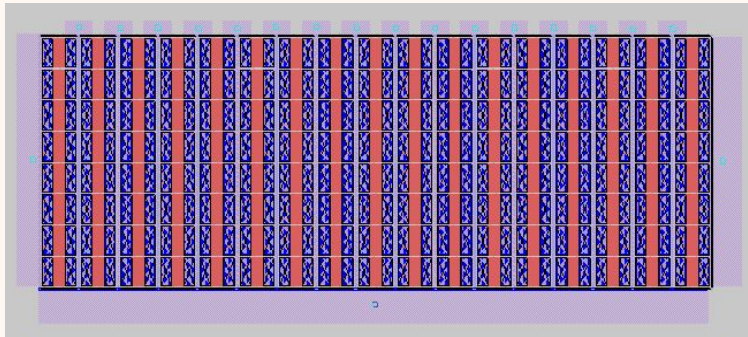
16 to 4 Priority Encoder

Flash ADC - Layout

PTAT

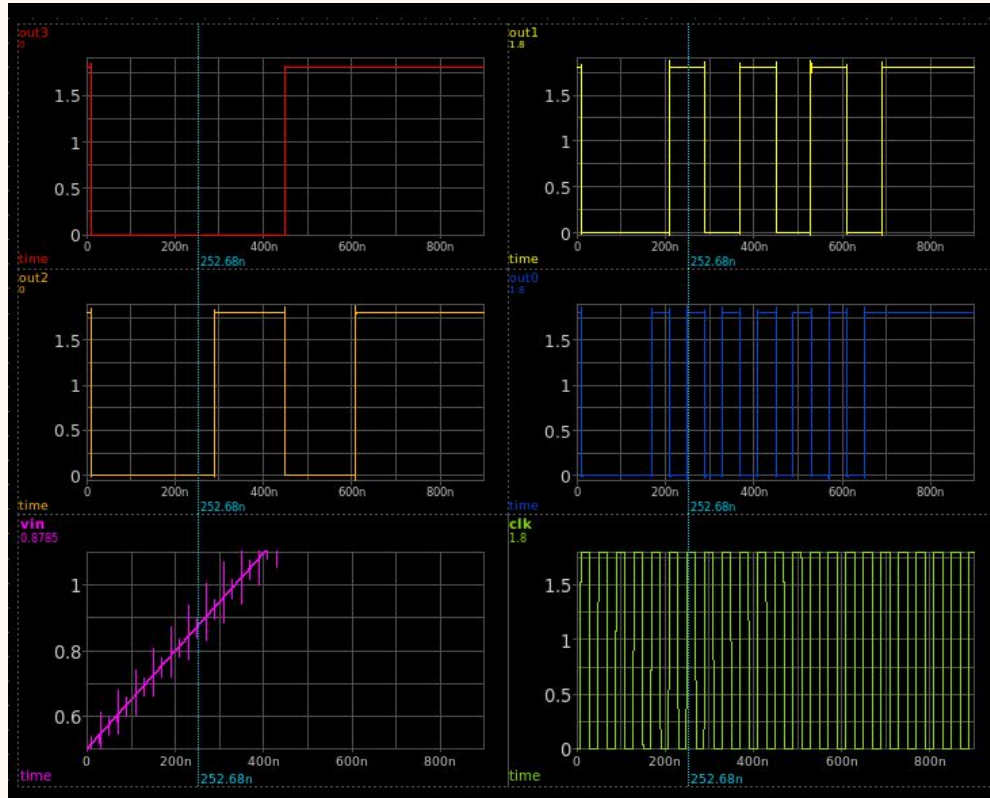


Resistor Ladder



Top Level
Flash ADC
Layout

Flash ADC – Post Extraction Simulations



- Post extraction sims correctly counting up with linear ramp
- Requires some tuning of VL and CLK speed
- Working on reducing potential capacitance and resistances
- Will run updated post-extraction simulations soon
- Comparisons with schematic level simulations will be made